

Magnets

MAGNETS PRODUCE A MAGNETIC FIELD, WHICH ATTRACTS SOME MATERIALS AND CAN ATTRACT OR REPEL OTHER MAGNETS.

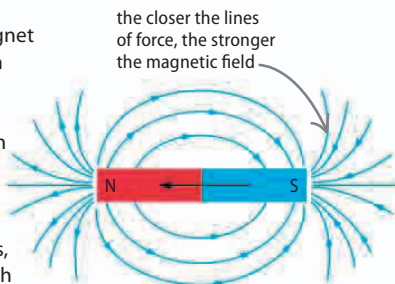
Some magnets occur naturally, while some materials can be made magnetic by passing an electric current through them. Some materials can be permanently magnetized.

Magnetic force

In magnetic materials, areas called domains behave like tiny magnets. When not magnetized, these are all jumbled up and point in different directions, but when placed in a magnetic field or stroked repeatedly by a magnet, the domains all line up so that all their north poles point in one direction and the south poles in the opposite direction, making the material magnetic.

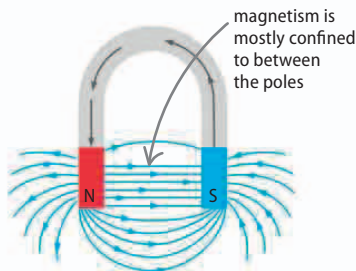
▷ Single bar magnet

The area around the magnet where its magnetism can affect other materials is called its magnetic field. A bar magnet has a north pole at one end and a south pole at the other. Cutting a bar magnet in two creates two magnets, each with their own north and south poles.



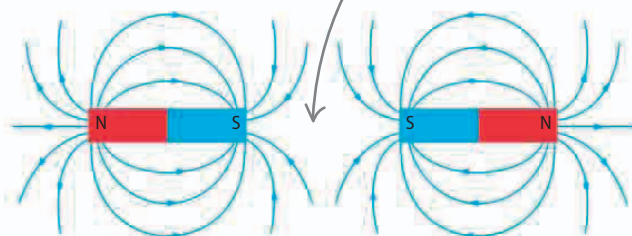
▷ Horseshoe magnet

Magnets come in all kinds of shapes, such as the horseshoe magnet. This type of magnet also has a north and south pole, but it is curved, so the poles are close together.



▽ Attract or repel

Two magnets will be attracted to each other if unlike poles (one north and one south) face each other. However, like poles repel, pushing each other away.



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Permanent magnets

Some materials, including iron, nickel, cobalt, and their alloys (metals combined with metals or nonmetals), are ferromagnetic. These can be magnetized by an electric current or by stroking another magnet. Once magnetized, these materials stay magnetic unless demagnetized by a shock, excess heat, or a variable magnetic field.

▽ Magnetic objects

Steel is an alloy of iron, and is used to make cans and paper clips. “Copper” coins actually contain nickel.

▽ Nonmagnetic objects

Common plastics are not magnetic, nor are aluminum beer and soda cans, or brass musical instruments.



REAL WORLD

Lodestone compass

Lodestone is a naturally occurring magnetic mineral that was used thousands of years ago to make the first compasses. If a piece of lodestone is allowed to spin freely, it will align itself with the Earth's magnetic field, pointing in a north–south direction. The word “magnet” comes from “Magnesia,” the area in Greece where lodestones and manesium were found.

